AN INTELLIGENT SYSTEM FOR AI-BASED PLAGIARISM DETECTION

Synopsis submitted in the partial fulfilment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

by

Submitted by

Jayant (2822666)
Deepak (2822667)
Ankit (2822637)
Umesh (2822642)

Under the Supervision of

Mrs. Aashima Rathi (Assistant Professor)



Department of Information Technology

PANIPAT INSTITUTE OF ENGINEERING & TECHNOLOGY SAMALKHA, PANIPAT 132102

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CANDIDATE'S DECLARATION

We hereby declare that the work which is being presented here entitled, **"An Intelligent System for AI-Based Plagiarism Detection"** by **Jayant, Deepak, Ankit, Umesh** in partial fulfillment of requirements for the award of the degree of **B.Tech.** in **Information Technology** submitted in the Department of Information Technology, PIET, Panipat affiliated to Kurukshetra University, is an authentic record of our own work under the supervision of Mrs. Aashima Rathi.

Signature of the Students Date: 28.01.2026

Name of Students: Jayant, Deepak, Ankit, Umesh

Roll Numbers: 2822666, 2822667, 2822637, 2822642

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Mrs. Aashima Rathi
Assistant Professor
PIET, Panipat

Signature of HOD
(Dr. Suman Mann)
Professor & Head, Department of Information Technology

ABSTRACT

Academic integrity is the cornerstone of the educational system, yet the ease of access to digital content and Generative AI tools has made plagiarism—both human and machine-generated—a critical challenge. Traditional plagiarism checkers primarily rely on string-matching algorithms, which often fail to detect paraphrased content or text generated by Large Language Models (LLMs) like ChatGPT.

To address this issue, this project proposes an "AI-Based Intelligent Plagiarism Detection System". The system integrates Natural Language Processing (NLP), Semantic Analysis, and Machine Learning to perform a comprehensive originality check.

For Text Similarity, the system utilizes Cosine Similarity and Jaccard Similarity on vector embeddings to detect direct copying. For Semantic Analysis, it employs BERT (Bidirectional Encoder Representations from Transformers) to identify paraphrasing where words are changed but the meaning remains the same. Additionally, an AI-Text Detection module is incorporated to flag content likely generated by AI models.

The application is implemented as a web-based system using the Flask framework, providing a user-friendly interface for document uploads (PDF, DOCX). The proposed system is cost-effective, real-time, and scalable, aiming to promote fairness and critical thinking in education.

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1. Introduction of the Problem and Objectives

Project Title Justification The project title **"AI-Based Intelligent Plagiarism Detection System"** aptly captures the essence of the main theme of ensuring academic integrity through plagiarism detection. The word **"AI-Based"** in the project title emphasizes the system's ability to perform beyond mere text similarity analysis by **using Natural Language Processing (NLP) and Semantic Analysis** to identify paraphrasing and AI-written content. The project title is specific, accurate, and relevant to the field of educational technology.

General Introduction to the Problem Academic integrity is the foundation of education. But with the rapid increase in digital content and the availability of Generative AI tools such as ChatGPT, plagiarism has moved from mere "copy-pasting" to sophisticated paraphrasing and AI-authored fabrication.

- Existing traditional plagiarism checkers often rely on simple string-matching algorithms, which fail to detect paraphrased content or translated text.
- Many current solutions are expensive (e.g., Turnitin) or lack real-time processing capabilities for students and small institutions.
- This project addresses these challenges by developing a web-based intelligent system capable of detecting both verbatim plagiarism and semantic similarities using advanced NLP models.

Specific Objectives The project aims to achieve the following:

- Develop a **text similarity detection engine** using Cosine Similarity and Jaccard Similarity.
- Implement **semantic analysis** using **BERT (Bidirectional Encoder Representations from Transformers)** to detect paraphrased content.
- Integrate **AI-generated text detection** to identify content created by Large Language Models (LLMs).
- Build a user-friendly **web-based application** using Flask/Django for easy document upload and reporting.
- Provide a scalable system that generates a detailed similarity percentage report with source links.

Justification of Objectives This project deserves approval as it directly contributes to **Quality Education (SDG 4)** by promoting honesty and fairness in assessments. Unlike traditional keyword-matching systems, this project uses **semantic embeddings**, making it robust against students who try to "trick" the system by changing a few words. From an academic perspective, it offers hands-on experience in **NLP, Vector Embeddings, and Full-Stack Web Development**.

Strengths and Limitations

- **Strengths:**
    - Detects semantic (meaning-based) plagiarism, not just exact matches.
    - Open-source and cost-effective compared to enterprise solutions.
    - Web-based and platform-independent.
- **Limitations:**
    - Processing speed may decrease with extremely large datasets or internet-wide scanning.
    - AI-detection accuracy can vary depending on the "temperature" of the generated text.

2. Literature Review

R. Rastgoo, K. Kiani, and S. Escalera [1] conducted a comprehensive survey on deep learning approaches. Their work reviews state-of-the-art techniques, emphasizing that **Deep Neural Networks (DNNs) and Recurrent Neural Networks (RNNs)** have significantly outperformed traditional statistical methods in pattern recognition. While their study focused on gesture recognition, the fundamental finding—that deep learning handles variations in data better than static algorithms—is directly applicable to identifying linguistic patterns in plagiarized text.

The International Journal of Computer Engineering [3] published research utilizing **BERT and LSTM networks** for recognition tasks. This approach combines natural language processing with temporal sequence modeling, demonstrating that transformer-based architectures enhance accuracy by capturing **contextual relationships** between data points. In the context of plagiarism, this technology is crucial for identifying "semantic plagiarism," where words are replaced with synonyms but the sentence structure and meaning remain identical.

A. Vaswani et al. [5] introduced the **Transformer architecture** with attention mechanisms that have revolutionized sequence learning. Key to this research is the **"Self-Attention"** mechanism, which allows the model to weigh the importance of different words in a sentence regardless of their distance. Unlike older models that process text linearly, the Transformer reads entire sentences at once, making it well-suited for detecting complex paraphrasing where word order is manipulated.

A. Radford et al. [6] presented work on **Large-Scale Weak Supervision**, demonstrating effective strategies for training models on massive, noisy datasets. This research underpins the functionality of Large Language Models (LLMs) like GPT. Understanding these architectures is essential for the **"AI-Detection"** module of this project, as it allows the system to analyze statistical properties (such as perplexity) to distinguish between organic human writing and synthetic AI-generated text.

The World Health Organization [7] and other global bodies emphasize the importance of standardization and accessibility. In the context of this project, this establishes the need for a universally accessible, web-based tool that promotes **academic integrity** and fairness in education, ensuring that assessment standards are maintained globally.

The following table summarizes the key research focused on in this project:

Sr. No.	Author(s)	Research Focus	Technology Used	Outcomes	Limitations
[1]	R. Rastgoo, K. Kiani, S. Escalera	Deep learning approaches for pattern recognition	CNNs and RNNs	Proved that deep learning handles variations in data better than static algorithms.	Computational complexity is high; earlier RNNs struggle with very long documents.
[3]	IJCESEN	Contextual recognition using Transformers	BERT + LSTM networks	Enhanced accuracy by capturing contextual relationships (semantics) between data points.	Dataset limitations often lead to overfitting; requires fine-tuning for specific domains.
[5]	A. Vaswani et al.	Transformer architecture with attention mechanisms	Attention-based neural networks	Revolutionized sequence learning; captures long-range dependencies effectively.	The base model is general-purpose; requires adaptation (embeddings) for specific similarity tasks.
[6]	A. Radford et al.	Large-scale weak supervision for generative models	Transfer Learning / Weak Supervision	Demonstrated effective strategies for training models on massive, noisy datasets.	Detection requires distinguishing "perfect" machine text from human imperfection

Sr. No.	Author(s)	Research Focus	Technology Used	Outcomes	Limitations
[7]	WHO / Global Standards	Documentation of global standards and needs	Statistical Analysis	Establishes the social and academic importance of maintaining integrity standards.	Does not provide technical solutions, only contextual background.

3. Methodology and Workflow

Requirement Analysis and Research (Week 1)

- Study existing plagiarism detection algorithms (Rabin-Karp, Winnowing).
- Analyze datasets for AI-generated vs. Human-written text.

System Design and Environment Setup (Week 2)

- Configure **Flask/Django** backend and project structure.
- Set up NLP libraries (**spaCy, NLTK, HuggingFace Transformers**).

Core Logic Development (Week 3–5)

- **Preprocessing Module:** Tokenization, stop-word removal, and lemmatization.
- **Vectorization:** Implement TF-IDF and BERT embedding generation.
- **Similarity Engine:** Calculate Cosine Similarity scores between input text and database.

AI-Detection & Reporting (Week 6–7)

- Train/Load a classifier model (e.g., RoBERTa base) to flag AI-generated content.
- Generate PDF reports highlighting plagiarized sections.

Testing and Optimization (Week 8–9)

- Validate accuracy using a test dataset of known plagiarized documents.
- Optimize processing time for large files.

User Interface and Deployment (Week 10)

- Design interactive web interface for file upload (Drag & Drop).
- Final documentation and demonstration.

(Include Data Flow Diagrams here similar to Fig 1 and Fig 2 in your file, but replacing "Sign Language" with "Document Processing" flow).

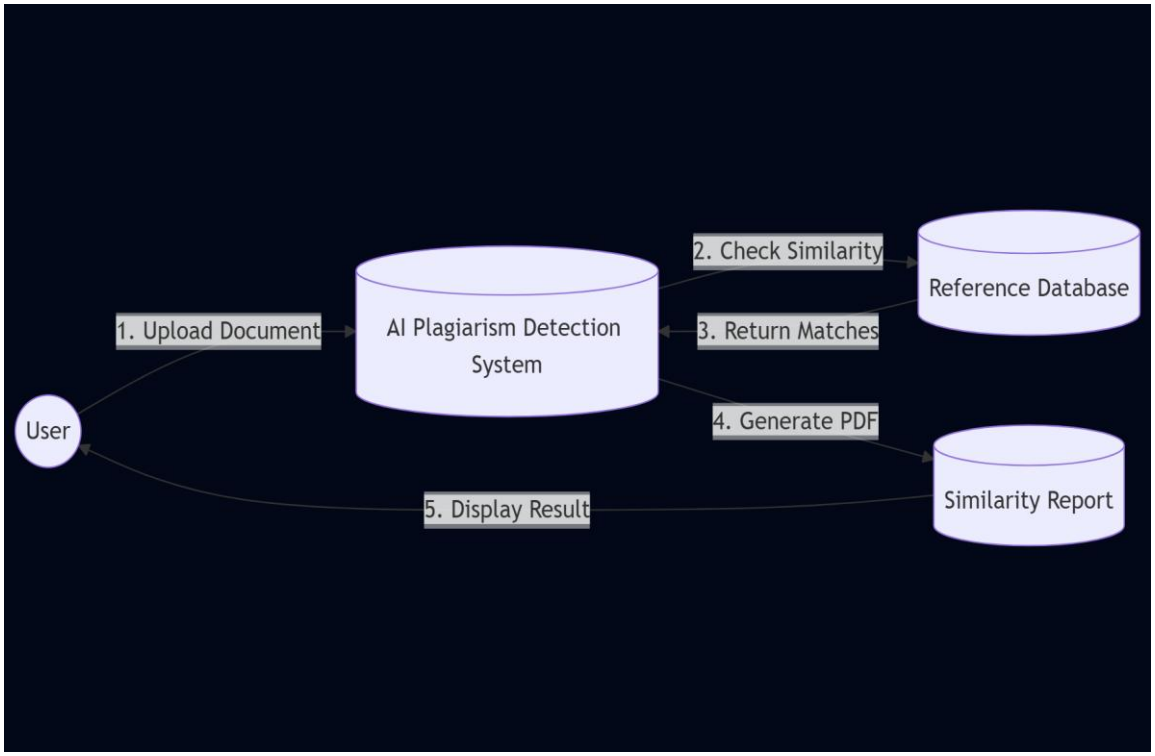


Fig 1. Level 0 Data flow Diagram

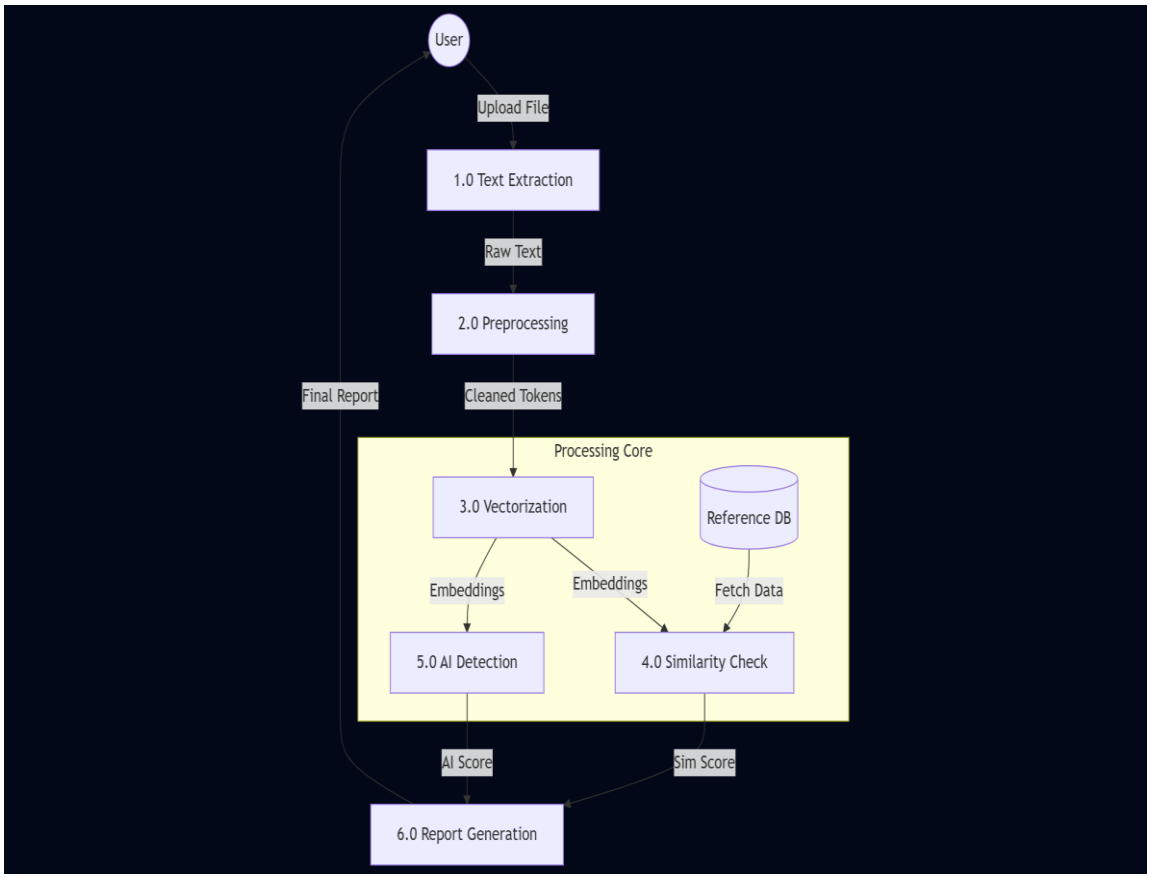


Fig 2. Level 1 Data Flow Diagram

4. Hardware and Software Requirements

Hardware Requirements:

- **Processor:** Intel Core i5 or higher (for model inference).
- **RAM:** 8GB or more (16GB recommended for BERT models).
- **Storage:** 256GB SSD or more.
- **GPU:** Optional (NVIDIA GTX/RTX recommended for faster NLP processing).

Software Requirements:

- **OS:** Windows 10/11, Ubuntu, or macOS.
- **IDE:** Visual Studio Code / PyCharm.
- **Languages:** Python (Core Logic), HTML/CSS/JS (Frontend).
- **Libraries:** Flask/Django, TensorFlow/PyTorch, Scikit-learn, NLTK, Transformers.

5. Project Categorization

- **Application-based:** The project is aimed at developing a software application to solve the specific problem of academic dishonesty.
- **Research-based:** The project involves the study, implementation, and fine-tuning of advanced **Natural Language Processing (NLP)** algorithms (such as BERT and Cosine Similarity) to solve the complex challenge of detecting semantic paraphrasing and AI-generated content.

6. Project Map with SDG Goals

The Project aligns with the following SDG goals:

- **SDG 4: Quality Education** – Ensures fair assessment and promotes original critical thinking among students.
- **SDG 9: Industry, Innovation, and Infrastructure** – Utilizes cutting-edge AI technology to automate and improve educational infrastructure.

7. Conclusions

The **“AI-Based Intelligent Plagiarism Detection System”** introduces an innovative automated assessment tool that bridges the gap between traditional manual verification and modern semantic understanding. Unlike legacy plagiarism checkers that focus solely on exact one-way string matching or require expensive enterprise licenses, this project emphasizes **affordability, real-time processing, and software-based intelligence** using advanced Natural Language Processing (NLP).

By leveraging **Flask**, **Deep Learning models (BERT)**, and **Vector Embeddings**, the system provides an interactive platform for converting raw text into detailed similarity metrics. It successfully addresses the dual challenge of modern education: detecting traditional "copy-paste" plagiarism and identifying synthetic content generated by AI tools like ChatGPT.

The modular architecture of the system allows for significant **future enhancements**, such as:

- **Cross-Lingual Detection:** Comparing documents across different languages (e.g., detecting if English text was translated from Hindi).
- **Smart Writing Suggestions:** Evolving from a "policing" tool to a "tutoring" tool that offers students constructive suggestions on how to rephrase flagged content originally.
- **OCR Integration:** Extending the system to scan handwritten assignments and physical textbooks.

The positive impact of this project spans multiple domains:

- **For Academics:** It serves as a comprehensive educational platform for students to understand complex concepts such as **Machine Learning**, **Vector Space Models**, and **Full-Stack Web Development**. It transforms theoretical NLP knowledge into a practical, real-world application.
- **For Colleges:** The system can be seamlessly integrated into existing Learning Management Systems (LMS) or computer science curricula. It provides institutions with a cost-effective, self-hosted solution to automate preliminary grading checks, saving valuable faculty time.

8. Specific Contributions of Team Members

Jayant (Team Leader)

- **Role:** Team Leading and Backend Development.
- **Leadership:** Oversees overall project planning, progress tracking, and ensures milestones and deadlines are achieved.
- **Backend Development:** Designs and manages the backend architecture, routing, API integration, and system logic.
- **System Logic:** Implements the core logic for the similarity algorithms (Cosine/Jaccard) and NLP processing models.
- **Review:** Reviews, tests, and validates all modules to ensure smooth integration and reliable system performance.

Deepak (Backend Developer)

- **Role:** Backend Development and System Support.
- **Backend Logic:** Assists in developing server-side functionalities using Flask/Django.
- **Data Handling:** Manages data flow between frontend, backend, and the AI models.
- **System Integration:** Supports integration of file processing, text extraction, and database management.

Ankit (Frontend Developer and Research)

- **Role:** Frontend Design and Research Analysis.
- **Frontend Implementation:** Designs and develops the user interface for document upload and result visualization.
- **Research:** Conducts research on existing plagiarism detection systems and NLP techniques.
- **User Experience:** Ensures intuitive interaction for text input and report display.

Umesh (Research and Documentation)

- **Role:** Research and Documentation.
- **Documentation:** Prepares project reports, documentation, and presentation materials.
- **Requirements:** Identifies system requirements and limitations.
- **Testing Support:** Assists in usability testing and interface optimization.
- **Analysis:** Analyzes datasets required for testing the similarity models.

9. Key References (In IEEE format only)

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